Comput Stat
DOI 10.1007/s00180-017-0732-4

ORIGINAL PAPER

Some hypothesis tests based on random projection

Ricardo Fraiman[1] · **Leonardo Moreno**[2] · **Sebastian Vallejo**[3]

Received: 19 August 2016 / Accepted: 27 April 2017
© Springer-Verlag Berlin Heidelberg 2017

Abstract Two new non-parametric tests are proposed based on continuous one-dimensional random projections. The first one addresses central symmetry and the second addresses independence. These tests are implemented for finite and infinite dimensional (functional) data sets. Both tests are distribution-free and universally consistent. Additionally, different techniques are proposed to improve the power of the tests. Promising results have been obtained by comparing the new tests with existing ones using simulation study. Real data in Banach spaces have been used to develop an application.

Keywords Functional data · Random projections · Central symmetry · Independence tests · Universal consistency

✉ Leonardo Moreno
mrleo@iesta.edu.uy

Ricardo Fraiman
rfraiman@cmat.edu.uy

Sebastian Vallejo
svallejo@grupomas.software
http://www.grupomas.software

1 Facultad de Ciencias, Universidad de la República, Montevideo, Uruguay

2 Facultad de Ciencias Ecónomicas y de Adminitración, Universidad de la República, Montevideo, Uruguay

3 División Analytics, Grupo MAS, Montevideo, Uruguay

Published online: 13 May 2017

1 Introduction

In recent years many theoretical and applied statistical tools for the analysis of functional and finite dimensional data have been developed. However certain issues have arisen with regard to the dimension of the space, and the extension of classical methods has encountered limitations.

The objective of the current research is to develop an easy-to-apply method of low computational complexity with sufficient theoretical background to enable the efficient analysis of finite- or infinite-dimensional spaces, to consider two main problems: *symmetry and independence tests* using random projections. In particular, if the space dimension is finite but high, the tests maintain a good performance.

The projection pursuit technique (see Friedman and Stuetzle (1981), Friedman and Tukey (1974)) has been used increasingly in the last decades as a way to address these problems. For a general overview of these topics see Jones and Sibson (1987). In particular Blough (1989) offers an application to symmetry tests. However, while this technique extends univariate procedures to the multivariate setting, it is highly sensitive to the dimensionality of the space and is supported by empirical results only.

One way to solve this problem is provided by Cuesta-Albertos et al. (2007). In that study, the solution is to work on the basis of a generalization of the Cramér–Wold theorem through random one-dimensional projections. The same authors propose an application to goodness of fit tests in Hilbert spaces, see Cuesta-Albertos et al. (2006). Later, Cuevas and Fraiman (2009) extends the results to Banach spaces. Unlike the pursuit technique, this approach is based on the fact that to determine a probability measure on a multivariate or infinite dimensional space, it is enough to know only the distribution of the measure of one projection under certain assumptions, which has been chosen randomly.

Multivariate symmetry is a crucial notion in statistics, especially in non-parametric statistics; in multivariate location problems, for instance all non-parametric testing procedures require some concept of multivariate symmetry. In the case of spherical symmetry procedures are proposed by Brandwein and Strawderman (1991). A procedure is proposed for the weaker assumption of elliptical symmetry Hallin and Paindaveine (2002). Testing for central symmetry has been studied in several papers, e.g. Sen and Puri (1967), Ley (2011), Aki (1993), Neuhaus and Zhu (1998), Einmahl and Gan (2016) and Heathcote et al. (1995).

The importance of the independence assumption for inference arises, for example, in clinical studies using diseased subjects assumed to be independent in the population study; see for instance Albert et al. (2001).

Some methods such as Wilks (1935) and Puri and Sen (1971) are based on normality assumptions, or are not applyable if the dimension exceeds the sample size. A further limitation of multivariate extensions of methods based on ranks is that they are ineffective for testing non-monotone types of dependence; see Székely et al. (2007).

This paper is organized as it follows. Section 2 presents useful concepts and results that are used throughout the manuscript. First, the concepts of symmetry and independence are briefly discussed. Next, a characterization through one-dimensional projections is described for central symmetry and independence. In Sect. 3 a characterization through random projections is proposed for the symmetry and independence

problems of random elements. Sections 4 and 5 describe procedures to test central symmetry and independence respectively, based on random projections. Section 6 is dedicated to Monte Carlo experiments, and Sect. 7 presents the results from applying the proposed independence test to a real data set. All proofs are given in the Appendix.

2 Basic concepts and results

Below is a description of the basic definitions of independence and symmetry, in particular central symmetry, and some of the main concepts that will be used throughout the paper.

2.1 Symmetry and independence

Definition 1 *(Independent random elements)* Let X and Y be two random elements defined on a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ taking values in a Banach space E. $\mathbf{X}$ and $\mathbf{Y}$ are independent if and only if $\mathbf{X}^{-1}(A)$ and $\mathbf{Y}^{-1}(B)$ are independent events for all open sets $A, B \in E$.

There are several types of symmetry definitions; they can be found, for instance in Serfling (2006) along with their main properties. Some are briefly described below.

Definition 2 *(Spherical Symmetry)* A random vector $\mathbf{X} \in \mathbb{R}^d$ has a spherically symmetric distribution around the origin if and only if $\mathbf{X}$ and $A\mathbf{X}$ have the same distribution for all orthogonal matrices $A_{d\times d}$.

Definition 3 *(Elliptical Symmetry)* A random vector $\mathbf{X} \in \mathbb{R}^d$ has an elliptically symmetric distribution around the origin if and only if $\mathbf{X}$ and $A^{'}\mathbf{Y}$ have the same distribution for some matrix $A_{k\times d}$ such that $A^{'}A = \Sigma$ with $rank(\Sigma) = k \leq d$ and for some $\mathbf{Y} \in \mathbb{R}^k$ having a spherical distribution with respect to the origin. The transposed matrix of A is denoted by $A^{'}$

The following definition of central symmetry also holds for infinite dimensional spaces.

Definition 4 *(Central Symmetry)* Let E be a Banach space. A random element $\mathbf{X}$ from E has a centrally symmetric distribution around the origin if and only if $\mathbf{X}$ and $-\mathbf{X}$ have the same distribution.

Observe that central symmetry is the weakest of the three concepts.

2.2 A characterization through one-dimensional projections

In order to develop tests for symmetry and for independence based on random projections we need as a first step to characterize both properties in terms of the one–dimensional projections of the underlying distribution. More precisely, we will see that symmetry and independence properties can be completely determined through

one-dimensional projections of the underlying distribution. Indeed, if π_h denotes the orthogonal projection of $\mathbb{R}^d$ onto the subspace spanned by h (with norm one), and B is a Borel set in this subspace, the induced measure on the subspace is given by

$$P_{\langle h \rangle}(B) = \mathbb{P}\left[\pi_h^{-1}(B)\right]. \tag{1}$$

The following set will be crucial for the approach. Let $\mathbb{P}$ and $\mathbb{Q}$ be Borel measures on $\mathbb{R}^d$. Then the following set is defined

$$\mathcal{E}(\mathbb{P}, \mathbb{Q}) = \{h \in \mathbb{R}^d / P_{\langle h \rangle} = Q_{\langle h \rangle}\}. \tag{2}$$

First, note that one can rewrite the Cramér–Wold theorem Cramér and Wold (1936) in terms of the set $\mathcal{E}(\mathbb{P}, \mathbb{Q})$ by

$$\mathcal{E}(\mathbb{P}, \mathbb{Q}) = \mathbb{R}^d \Leftrightarrow \mathbb{P} = \mathbb{Q}. \tag{3}$$

For an infinite dimensional Banach space E, the set $\mathcal{E}(\mathbb{P}, \mathbb{Q})$ given in Eq. (2) must be redefined using elements of the dual space E^* to replace the inner product in the definition. More precisely, if $\mathbf{X}$ and $\mathbf{Y}$ are random elements in a separable Banach space E with distributions $\mathbb{P}$ and $\mathbb{Q}$ respectively , and E^* stands for the dual space we define

$$\mathcal{E}(\mathbb{P}, \mathbb{Q}) := \mathcal{E}(\mathbf{X}, \mathbf{Y}) = \{f \in E^* / f(\mathbf{X}) \text{ and } f(\mathbf{Y}) \text{ have the same distribution}\}. \tag{4}$$

A functional version of the Cramér–Wold theorem can therefore be stated as (see Padgett and Taylor (1973))

$$\mathcal{E}(\mathbf{X}, \mathbf{Y}) = E^* \Leftrightarrow \mathbf{X} \text{ and } \mathbf{Y} \text{ have the same distribution} \Leftrightarrow \mathbb{P} = \mathbb{Q}. \tag{5}$$

Based on these facts, central symmetry and independence of random elements can be characterized in terms of the distributions of one-dimensional linear functions. Indeed, from the linearity of f, the functional version of the Crámer–Wold Theorem (5) and Lemma 2.3.5 in Padgett and Taylor (1973) we can derive easily the following two results:

Lemma 1 (Characterization of Central Symmetry) *Let $\boldsymbol{X}$ be a random element in a Banach space E. $\boldsymbol{X}$ is centrally symmetric if and only if $f(\boldsymbol{X})$ and $-f(\boldsymbol{X})$ are random variables with the same distribution for any $f \in E^*$*

Therefore, if all univariate projections of $\mathbf{X}$ are symmetric random variables, $\mathbf{X}$ has central symmetry.

Lemma 2 (Characterization of Independence) *Let E be a separable Banach space. Two random elements $\boldsymbol{X}$ and $\boldsymbol{Y}$ in E are independent if and only if $f(\boldsymbol{X})$ and $g(\boldsymbol{Y})$ are independent for all $f, g \in E^*$.*

For the simpler case of Hilbert spaces $(\mathcal{H}, \langle \cdot, \cdot \rangle)$ (including the finite dimension case) we can use the Riesz representation theorem that characterizes the elements of the dual space by inner products, i.e., for all $f \in \mathcal{H}^*$, $\exists h \in \mathcal{H}$, $\|h\| = 1$, such that

$f(x) = \langle h, X \rangle$. The characterization is through orthogonal projections of the random vectors onto all the possible directions in $\mathcal{H}$.

3 Random projections

In this section symmetry and independence will be characterized through random projections. The finite dimensional case will be presented first. We start a quick review of some results for random projections.

Carleman's condition. The following Carleman's condition is a key assumption in order to apply the results of random projections. Let P a Borel probability measure on $\mathbb{R}^d$ such that the absolute values of its moments $m_n = \int \|x\|^n P(dx)$ are finite. P satisfies the *Carleman's condition* if $\sum_{n\geq 1} m_n^{-1/n} = \infty$. If P satisfies the Carleman's condition then it is determined by its moments; see for instance Shohat and Tamarkin (1943).

Definition 5 S is a projective hyper-surface in $\mathbb{R}^d$ if and only if there exists a homogeneous polynomial $p(x)$ in $\mathbb{R}^d$ such that

$$S = \{x \in \mathbb{R}^d / p(x) = 0\}. \tag{6}$$

The following result (Theorem 1, in Cuesta-Albertos et al. (2007)) is key in what follows. This theorem allows to characterize a distribution by a one-dimensional projection chosen at random.

Theorem 1 (Cuesta, Fraiman and Ransford, 2007) *Let P and Q be Borel measures in $\mathbb{R}^d$, where $d \geq 2$. If the following holds*

- *P is determined by its moments,*
- *$\mathcal{E}(P, Q)$ is not contained in a projective hyper-surface in $\mathbb{R}^d$,*

then $P = Q$.

In particular, if the set $\mathcal{E}(P, Q)$ has a positive H-measure in $\mathbb{R}^d$, H being an absolutely continuous measure with respect to Lebesgue measure, this set is not contained in any projective hyper-surface. In Cuevas and Fraiman (2009) the former statement is generalized to random elements defined in separable Banach spaces E.

Let $\mathbb{P}$ be a probability measure on E and $f \in E^*$. The univariate distribution P_f is defined by

$$P_f(A) = \mathbb{P}\left(f^{-1}(A)\right),$$

for all A a Borel set of $\mathbb{R}$.

Theorem 2 (Cuevas and Fraiman, 2009) *Let μ be a Radon Gaussian non-degenerate measure on E^*. Let Q and M be probability measures on E such that*

- *the absolute moments $m_n = \int \|x\|^n dM(x)$ are finite and verify Carleman's condition,*
- *the set $\mathcal{E}(M, Q) = \{h \in E^* / Q_h = M_h\}$ has a positive μ-measure,*

then $Q = M$.

3.1 The symmetry and independence problems

From the previous results, conditions for the symmetry and independence problems of random elements will be derived. Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space and $\mathbf{X}$ a random element taking values in a Banach space E. We say that the absolute moments of $\mathbf{X}$ are finite if the absolute moments of the induced probability measures $M = \mathbb{P} \circ \mathbf{X}^{-1}$ are finite.

Theorem 3 *Let μ be a Gaussian non-degenerate Radon measure on E^*. Let $\boldsymbol{X}$ be a random element in E such that the following hold*

- *the absolute moments of $\boldsymbol{X}$ are finite and verify Carleman's condition,*
- *the set $\mathcal{E}(\boldsymbol{X}) = \{f \in E^* / f(\boldsymbol{X})$ is a symmetric random variable$\}$ and has a positive μ-measure,*

then $\boldsymbol{X}$ is a centrally symmetric random element.

Theorem 4 *Let $(\Omega, \mathcal{B}, \mathbb{P})$ be a probability space, $\boldsymbol{X}$ and $\boldsymbol{Y}$ are two random elements taking values in a separable Banach space E, and μ be a Radon non-degenerate Gaussian measure on $(E \times E)^*$. Assume that the absolute moments of $\boldsymbol{X}$ and $\boldsymbol{Y}$ are finite and that the following series diverges*

$$\sum_{n \geq 1} \min \left\{ m_{\boldsymbol{X}}^{-1/n}(n), m_{\boldsymbol{Y}}^{-1/n}(n) \right\} = \infty. \tag{7}$$

If the set

$$\mathcal{E}(\boldsymbol{X}, \boldsymbol{Y}) = \{h \in (E \times E)^* / h(\boldsymbol{X}, 0) \text{ and } h(0, \boldsymbol{Y}) \text{ are random independent variables}\}$$

has a positive μ-measure, then $\boldsymbol{X}$ and $\boldsymbol{Y}$ are independent.

Example 1 Consider a probability measure on $\mathbb{R}^3$ that is uniformly distributed in a ball of center $(0, 0, 1)$ and radius 1. Clearly, the distribution is not centrally symmetric with respect to the origin, but the projections over any direction in the plane xy are symmetric. However, the plane has zero Lebesgue measure in $\mathbb{R}^3$.

Example 2 On the other hand, if a case of perfect dependence is considered, when both random vectors are the same

$$\mathbf{X} = \mathbf{Y} = (X_1, X_2) \sim N\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right]. \tag{8}$$

$\mathbf{X}$ and $\mathbf{Y}$ are projected onto two unit vectors $u = (u_1, u_2)$ and $v = (u_3, u_4)$, and the projected variables are independent if and only if u and v are orthogonal. So

$$\begin{aligned} \mathcal{E}(\mathbf{X}, \mathbf{Y}) &= \{(u, v) \in \mathbb{R}^4 / \langle \mathbf{X}, u \rangle \text{ and } \langle \mathbf{Y}, v \rangle \text{are indep.}\} \\ &= \{(u_1, u_2, u_3, u_4) \in \mathbb{R}^4 / u_1 u_3 + u_2 u_4 = 0\}, \end{aligned}$$

and while the set of pairs of directions is infinite, these are contained in a hyper-projective surface of degree 2 in $\mathbb{R}^4$, which also has Lebesgue measure zero.

4 A central symmetry test

Let $\{\mathbf{X}; \mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n\}$ be a set of i.i.d. random elements in a separable Banach space E whose distribution is determined by its moments. To perform a central symmetry test in E, namely

$$\begin{aligned} &H_0)\ \mathbf{X} \text{ and } -\mathbf{X} \text{ have the same distribution,} \\ &H_1)\ \mathbf{X} \text{ and } -\mathbf{X} \text{ do not have the same distribution,} \end{aligned} \tag{9}$$

one may apply the following procedure:

- $h \in E^*$ is randomly drawn with a Gaussian μ-measure defined as in Theorem 3. For the finite dimensional case, it suffices to draw a direction h generated by a measure H on $\mathbb{R}^d$ (H being absolutely continuous with respect to Lebesgue measure). In both cases take $\|h\| = 1$.
- For a fixed h, build a sample of the i.i.d. random variables induced by h,

$$\{h(\mathbf{X}_1), h(\mathbf{X}_2), \ldots, h(\mathbf{X}_n)\}. \tag{10}$$

  For finite dimensional spaces, the i.i.d. sample $\{\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n\}$ is orthogonally projected onto the one-dimensional space generated by h, thereby obtaining a sample of random variables in $\mathbb{R}$. Write $h(\mathbf{X}) = \langle \mathbf{X}, h \rangle$.
- Given a significance level α, a one–dimensional symmetry test of the Kolmogorov–Smirnov class is performed with the projected data given in (10), as developed by Sen and Chatterjee (1973). More precisely, if F^h denotes the cumulative distribution function of the random variable $h(\mathbf{X}_1)$, the one-dimensional test addresses the problem

$$\begin{aligned} &H_0)\ \ F^h(x) + F^h(-x) - 1 = 0\ \forall x \in \mathbb{R}, \\ &H_1)\ \ |F^h(x) + F^h(-x) - 1| > 0 \text{ for some } x \in \mathbb{R}, \end{aligned} \tag{11}$$

  where for a given distribution function F, $F(x^-)$ stands for the one side limit $\lim_{t \to x^-} F(t)$. The proposed test statistic is given by

$$D^h(n) = \sup_{x \geq 0} |F_n^h(x) + F_n^h(-x^-) - 1| = \max\left[D_-^h(n), D_+^h(n)\right], \tag{12}$$

  with

$$\begin{aligned} D_-^h(n) &= \sup_{x \geq 0}\left(1 - F_n^h(x) - F_n^h(-x^-)\right) \textit{ and} \\ D_+^h(n) &= \sup_{x \geq 0}\left(F_n^h(x) + F_n^h(-x^-) - 1\right), \end{aligned}$$

  where F_n^h stands for the empirical distribution of the univariate data

$$\{h(\mathbf{X}_1), h(\mathbf{X}_2), \ldots, h(\mathbf{X}_n)\}.$$

For "large" values of the statistic H_0, reject the null hypothesis in (9). This test is denoted by RPK_1.

4.1 Exact distribution of $D^h(n)$ under H_0

Let $\mathcal{F}_0$ be the set of all symmetric and continuous distributions on E. We will now, as in Sen and Chatterjee (1973), determine the exact distribution of the statistic under H_0, verifying that it is distribution–free, i.e, its distribution neither depends on the measure H used to choose the direction h, nor on the distribution $F \in \mathcal{F}_0$ of the data.

Theorem 5 [Distribution of the test statistic under H_0] *For all $F \in \mathcal{F}_0$ and any "direction" $h \in E^*$, the following holds*

$$P\left(nD_-^h(n) \geq k\right) = P\left(nD_+^h(n) \geq k\right) = 2\sum_{i=0}^{s}\binom{n}{i}2^{-n} - \delta_k\binom{n}{s}2^{-n}, \tag{13}$$

where $s = [n/2k] - 1$, $k = 1, 2, \ldots, n$ and δ_k is 0 if $n-k$ is even but 1 if $n-k$ is odd. Thus,

$$P\left(nD^h(n) \geq k\right) = \begin{cases} 1 & \text{if } k = 0, 1 \\ 2\sum_{j=0}^{u}(-1)^j P\left[nD_-^h(n^-) \geq (2j+1)k\right] & \text{if } k > 1, \end{cases} \tag{14}$$

where $u = [n/2k] - 1$.

It is also possible to obtain the asymptotic distribution of the statistic under the null hypothesis H_0.

4.2 Consistency of the test

In this section it is shown that the proposed test is universally consistent under any non-symmetrical alternative.

Theorem 6 (Consistency of the Test) *Let $\{X_n\}_{n\geq 1}$ be a sequence of i.i.d. random elements with distribution determined by its moments, such that X_1 is not centrally symmetric. So, having a measure H absolutely continuous with respect to Lebesgue measure on $\mathbb{R}^d$, it holds that*

$$H\left\{h \in \mathbb{R}^d : P\left(\liminf_{n\to+\infty} D_n^h > 0\right) = 1\right\} = 1.$$

4.3 Power of the test

We have already shown that just using a single random projection the proposed test is distribution–free under H_0 (it does not depend on the underlying distribution of

the data or on the measure H of the random projection) and is consistent under any alternative. However, for finite sample sizes, the test may have low power. This problem was already analyzed in Cuesta-Albertos et al. (2006). There are two reasons for this loss of power. First, although under H_1 the set of directions where the projection has a symmetric distribution has H-measure zero, when considering the empirical distribution instead of the true underlying distribution, it may happen that there is a small 'angular neighbourhood' of directions, with positive measure, where H_0 is not rejected, although it should be rejected. Second, univariate non-parametric tests of the Kolmogorov–Smirnov type do not have a high power for any alternative even when they are universally consistent. One solution for this second problem could be adjusting the statistic by modifying the original one as in Mason and Schuenemeyer (1983) for goodness of fit tests for testing symmetry. To overcome the first problem, we propose two different alternatives to make the test more robust, and we perform a small simulation study to evaluate the performance of these alternatives. The first one, also presented in Cuesta-Albertos et al. (2006) is to take a finite number of random directions, perform the univariate test for each direction, and then compute the maximum as the statistical test. The test based on j projections will be denoted RPK_j. The second one is by 'conveniently' choosing H, with the understanding that convenient means those distributions absolutely continuous with respect to Lebesgue measure which allocate greater probability mass to the set of directions where the distribution of the data could be more asymmetric. For each direction, an asymmetry index is defined as the absolute value of the median of the projected data in that direction. The distribution of the statistic is still free under H_0, and in this way, the power of the univariate test is increased while maintaining the significance level. That is, under H_0 the choice of the distribution H does not affect the error type 1 of the test. H is chosen in a data driven way, which tries to assign more probability mass to those directions where the distribution is asymmetric. The discussion will be focused on how to choose the asymmetry index and how to determine H. This test will be referred to as RPKW.

The next section will show that an improvement in power occurs when the significance level of each univariate test has to be reduced, that is, if one wants to control the significance level of the original test above a certain threshold. Bonferroni correction is used for RPKj and the distribution–free property under H_0 is lost.

5 An independence test

Let $\{(\mathbf{X}, \mathbf{Y}); (\mathbf{X}_1, \mathbf{Y}_1), (\mathbf{X}_2, \mathbf{Y}_2), \ldots, (\mathbf{X}_n, \mathbf{Y}_n)\}$ be a set of i.i.d. random elements in $E \times E$, and E be a separable Banach space, where the distribution of $(\mathbf{X}, \mathbf{Y})$ is determined by its moments. A test is proposed to contrast

$$\begin{aligned} &H_0)\ \mathbf{X} \text{ and } \mathbf{Y} \text{ are independent,} \\ &H_1)\ \mathbf{X} \text{ and } \mathbf{Y} \text{ are not independent.} \end{aligned} \tag{15}$$

The goal, as in the symmetry tests, is to extrapolate statistical methods that have been developed for finite dimensions. Mainly an independence test for infinite dimensional data based on copulas is used, whose theory was intrinsically conceived for finite dimensions. The results are obtained for the space $E \times E$, E being a separable Banach space, but it is easy to extend it to the case $E_1 \times \ldots \times E_k$.

Let $(\mathbf{X}, \mathbf{Y})$; $(\mathbf{X}_1, \mathbf{Y}_1), \ldots, (\mathbf{X}_n, \mathbf{Y}_n)$ be a sample of i.i.d. random elements in $E \times E$, it is required to determine whether $\mathbf{X}$ and $\mathbf{Y}$ are independent elements.

Under some conditions on the moments and choosing a non-degenerate Gaussian measure μ in the space $(E \times E)^*$ dual of $(E \times E)$, we will show that if the set

$$\mathcal{E}(\mathbf{X}, \mathbf{Y}) = \left\{h \in (E \times E)^* / h(\mathbf{X}, 0) \text{and} h(0, \mathbf{Y}) \text{are independent random variables}\right\},$$

has a positive μ-measure, μ being a Gaussian random measure on $(E \times E)^*$, then $\mathbf{X}$ and $\mathbf{Y}$ are independent. The proposed methodology is the following:

- Draw $h \in (E \times E)^*$ generated by μ. We define $f(x) = h(x, 0)$ and $g(y) = h(0, y)$. Observe that $f, g \in E^*$.
- For fixed f and g, an independence test from the empirical copula of the random variables $f(\mathbf{X})$ and $g(\mathbf{Y})$, $C_n^{f(\mathbf{X}), g(\mathbf{Y})}$, proposed by Genest et al. (2007) is implemented. It is a function of the empirical process associated with the copula

$$\sqrt{n}[C_n^{f(\mathbf{X}), g(\mathbf{Y})}(u, v) - u.v] \quad \text{such that} \quad (u, v) \in [0, 1]^2.$$

Its asymptotic distribution is computed under null hypothesis, whereas based on Fermaninan (2005) and Fermaninan et al. (2004) it follows that the test is consistent under any alternative.

5.1 Asymptotic distribution of the statistic under H_0 and its consistency

Set $f(x) = h(x, 0)$ and $g(y) = h(0, y)$. Draw from h the i.i.d. sample

$$\{(f(\mathbf{X}_1), g(\mathbf{Y}_1)), \ldots, (f(\mathbf{X}_n), g(\mathbf{Y}_n))\}.$$

We will use the statistic of the Crámer–von Mises type

$$W_n = \int_{[0,1]^2} \mathbb{C}_n^2(u, v) du dv, \tag{16}$$

with

$$\mathbb{C}_n(u, v) = \sqrt{n}[C_n^{f(X), g(Y)}(u, v) - u.v] \quad \text{such that} \quad (u, v) \in [0, 1]^2,$$

and $C_n^{f(X), g(Y)}(u, v)$ the empirical copula

$$C_n(u_1, u_2) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_{(R_{i1} \leq n u_1)} \mathbf{1}_{(R_{i2} \leq n u_2)},$$

with $R_{ij} = \sum_{k=1}^{n} \mathbf{1}_{(X_{kj} \leq X_{ij})}$ for $1 \leq i \leq 2$ and $j = 1, 2$.

In Fermaninan et al. (2004) some results of goodness of fit are presented for which this independence test is a particular case. It has been shown in Genest and Rémillard (2008), under some regularity hypotheses, that the test is universally consistent.

6 Simulations

6.1 Symmetry test simulations

The efficiency of the proposed test compared with other multivariate symmetry tests will be investigated. A comparison is performed through some simulations in $\mathbb{R}^d$ for $d = 2$, 3 and 50 for a sample of size $n = 100$, with the following tests:

1. (Test RPK_{10}.) Ten independent, uniformly distributed random directions in the unit sphere of $\mathbb{R}^d$ are chosen. Then, the sample is projected over each of the 10 random directions h, $< X_i, h >$. Next the p value is computed for each of the 10 projected univariate samples. If in any of these directions H_0 is rejected, the test rejects the null assumption. To compute the power this procedure is performed 10, 000 times, and the proportion of rejections is calculated. A rejection occurs when at least one of the ten univariate tests has a p value smaller than 0.005.
2. (Test RPK_1.) The same test as above but projecting in only one random direction and $\alpha = 0.05$.
3. (Test RPKW.) One hundred directions in the unit sphere are chosen. Next, the absolute value of the median of the projected data in each direction is computed, and the regression function is estimated, in which each direction of $\mathbb{R}^d$ has as an image it's absolute value median (by the use of cubic splines). Then, it is normalized to obtain a density that is used as the distribution H. A new direction is drawn again but now with distribution H, and the test is performed in this direction using $\alpha = 0.05$. An illustration is given in Fig. 1 where a sample of size 30 of a bivariate normal distribution $N(\boldsymbol{\mu} = (1/2, 1/2), \Sigma = \mathbf{I}_2)$ is generated and the absolute value of the median is computed for one hundred random directions. This function points to the directions where the data set is more asymmetric with respect to the

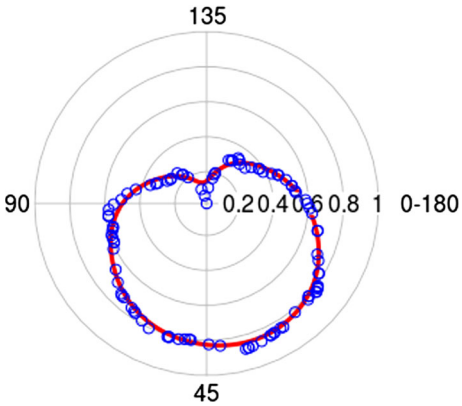

Fig. 1 Circular regression estimation of the absolute value of the median for one hundred directions

origin. Build up in this way the distribution H assigns greater probability to those directions (in this example, the directions close to 45 degrees.)

4. (Marden's Test.) Given a sample $\{X_1, \ldots, X_n\}$, the test developed by Marden in Marden (1999), first reordered the sample into $\{X_{A_1}, X_{A_2}, \ldots, X_{A_n}\}$ according to their increasing Euclidean distances to the origin. Then the following statistic $M^{(n)}$ is used to build up the test, rejecting the null hypothesis for large values of $M^{(n)}$,

$$M^{(n)} = \sqrt{\frac{2}{n}} \sum_{i=2}^{n} U'_{A_i} U_{A_{i-1}}, \tag{17}$$

where $U_{A_i} = X_{A_i}/\|X_{A_i}\|$. This test is asymptotically (standard) normally distributed under the null hypothesis of spherical symmetry around the origin.

5. (Ley's Test.) The test proposed by Ley (2011) is based on a depth measure D in $\mathbb{R}^d$ and the ordered sample $\{X_{A_1}, X_{A_2}, \ldots, X_{A_n}\}$ according to each observation depth in the symmetrized sample $\{\pm X_1, \pm X_2, \pm \ldots, \pm X_n\}$. The following statistic is proposed

$$R_D^{(n)} = 1 + \sum_{i=k+1}^{n} \mathbf{1}_{\{\mathbf{0} \in S(U_{A_i}, U_{A_{i-1}}, \ldots, U_{A_{i-k}})\}}, \tag{18}$$

where $S(x_1, x_2, \ldots, x_n)$ is the convex simplex defined by $(x_1, x_2, \ldots, x_n)$. Under the null hypothesis of central symmetry about the origin, along with some other assumptions, see Dyckerhoff et al. (2015), $n^{-1/2}(4R_D^{(n)} - n - 2)$ is asymptotically normal with mean zero and variance $\sigma^2 = 11/3$.

For the finite dimensional case two different scenarios developed for dimensions $d = 2$, 3 and 50 will be used. A third scenario is proposed for the infinite dimensional case. In Scenario 1 and Scenario 2 we want to test angular symmetry around the origin and in Scenario 3 we want to test angular symmetry around the null function.

- (Scenario 1.) An independent sample $\{X_1, \ldots, X_n\}$ with multivariate Student $t_4\left(\boldsymbol{\mu} = \mu \mathbf{1}_d, \Sigma = \mathbf{I}_d\right)$ distribution, where μ is a real number in [0; 0.5] and d is the spatial dimension.
- (Scenario 2.) An independent sample $\{X_1, \ldots, X_n\}$ of a skew normal multivariate distribution, see Azzalini and Valle (1996), $SN_d(\Sigma, \boldsymbol{\beta})$ with density

$$f(x) = 2\phi(x; \Sigma)\Phi(\boldsymbol{\beta}^t x) \quad x \in \mathbb{R}^d,$$

where $\phi(x; \Sigma)$ is the d-dimensional normal density with zero mean and correlation matrix Σ, $\Phi(\cdot)$ is the $N(0, 1)$ cumulative distribution function and $\beta \in \mathbb{R}^d$ is a skew parameter, see Fig. 2. We take $\Sigma = \mathbf{I}_d$ and $\boldsymbol{\beta} = \mu(1, 2, 1, \ldots) \in \mathbb{R}^d$, with μ taking values between 0 and 0.5.
- (Scenario 3.) For the infinite dimensional case a sample of size $n = 100$ is simulated from

$$X(t) = W(t) + \mu t, \tag{19}$$

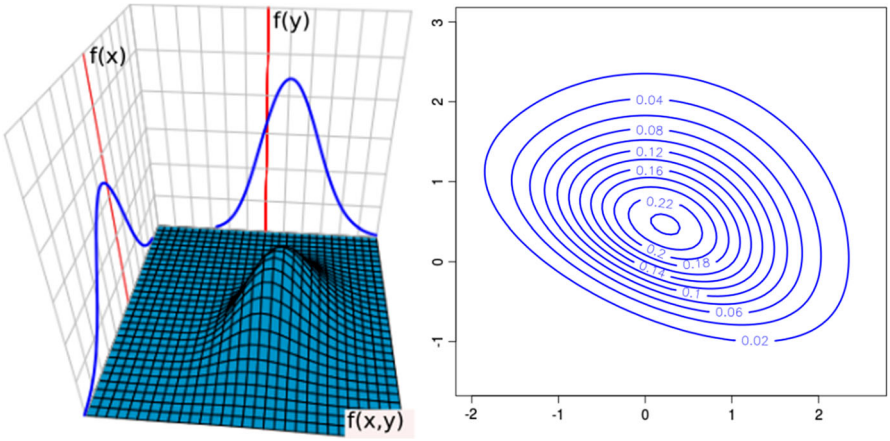

Fig. 2 Bivariate skew Normal with $\beta = (1, 2)$ and $\Sigma = \mathbf{I}_2$. *Left Panel* density plot f(x,y) and marginal densities plots f(x) and f(y). *Right Panel* level sets of f(x,y)

Table 1 Power of symmetry tests in Scenario 1 in dimensions 2 and 3

μ	Student in dim 2					Student in dim 3				
	RPK_1	Marden	Ley	RPKW	RPK_{10}	RPK_1	Marden	Ley	RPKW	RPK_{10}
0	0.05	0.05	0.06	0.05	0.03	0.04	0.05	0.05	0.05	0.04
0.05	0.07	0.08	0.08	0.07	0.04	0.09	0.05	0.09	0.06	0.06
0.10	0.10	0.09	0.10	0.12	0.09	0.14	0.06	0.11	0.16	0.15
0.15	0.11	0.10	0.11	0.31	0.22	0.21	0.10	0.12	0.31	0.34
0.20	0.14	0.14	0.13	0.54	0.41	0.35	0.14	0.17	0.63	0.58
0.25	0.21	0.28	0.22	0.83	0.59	0.44	0.21	0.18	0.67	0.78
0.30	0.30	0.37	0.31	0.92	0.80	0.51	0.31	0.25	0.70	0.91
0.35	0.38	0.41	0.38	0.94	0.90	0.58	0.48	0.32	0.75	0.97
0.40	0.52	0.54	0.52	0.95	0.96	0.62	0.60	0.45	0.82	0.99
0.45	0.62	0.66	0.68	0.98	0.99	0.69	0.74	0.66	0.90	1.00
0.50	0.76	0.77	0.80	0.99	1.00	0.74	0.79	0.70	0.97	1.00

where $W(t)$ is a Brownian motion in $[0, 1]$ and $\mu \in [0, 1/2]$.

The power functions of the different tests are given in Tables 1 and 2. It can be observed in all cases that the symmetry test with random projections outperforms its competitors.

The methods based on random projection RPK_{10} and RPKW have better results. While k-projection tests have powers similar to weighted projections, the latter is preferred because it retains the distribution-free property.

It is not possible to perform the Marden or Ley tests in dimension 50 due to the required computing time. Table 3 shows the power of RPK_1 and RPK_{10} in dimension 50 and in infinite dimension (the space $C(0, 1)$).

Table 2 Power of symmetry tests in Scenario 1 in dimensions 2 and 3

μ	Student in dim 2					Student in dim 3				
	RPK_1	Marden	Ley	RPKW	RPK_{10}	RPK_1	Marden	Ley	RPKW	RPK_{10}
0.00	0.05	0.04	0.05	0.05	0.03	0.05	0.05	0.06	0.05	0.03
0.05	0.12	0.05	0.07	0.05	0.07	0.07	0.06	0.07	0.06	0.04
0.10	0.23	0.07	0.08	0.20	0.20	0.16	0.10	0.08	0.12	0.14
0.15	0.38	0.10	0.16	0.43	0.40	0.29	0.15	0.12	0.40	0.38
0.20	0.53	0.20	0.20	0.55	0.67	0.39	0.18	0.19	0.53	0.61
0.25	0.62	0.30	0.31	0.63	0.84	0.48	0.29	0.23	0.59	0.83
0.30	0.70	0.37	0.40	0.73	0.95	0.55	0.38	0.33	0.71	0.91
0.35	0.73	0.49	0.55	0.85	0.98	0.57	0.50	0.44	0.80	0.97
0.40	0.74	0.63	0.66	0.99	0.99	0.62	0.58	0.51	0.88	0.99
0.45	0.77	0.72	0.76	0.99	1.00	0.63	0.67	0.58	0.93	0.99
0.50	0.78	0.79	0.83	1.00	1.00	0.66	0.70	0.69	0.97	1.00

Table 3 Power of symmetry tests in Scenarios 1 and 2 in dimension 50. Power of symmetry tests in Scenario 3

μ	Student in dim 50.		Skew normal in dim 50.		$X(t)$ in $C[0, 1]$.	
	RPK_1	RPK_{10}	RPK_1	RPK_{10}	RPK_1	RPK_{10}
0.00	0.04	0.05	0.05	0.05	0.05	0.05
0.05	0.07	0.10	0.06	0.06	0.06	0.06
0.10	0.12	0.27	0.08	0.09	0.11	0.10
0.15	0.20	0.53	0.28	0.27	0.19	0.19
0.20	0.30	0.78	0.33	0.38	0.30	0.30
0.25	0.38	0.91	0.45	0.50	0.43	0.46
0.30	0.45	0.96	0.52	0.59	0.57	0.61
0.35	0.50	0.98	0.53	0.70	0.69	0.74
0.40	0.56	0.99	0.59	0.78	0.79	0.86
0.45	0.61	1.00	0.60	0.85	0.87	0.93
0.50	0.64	1.00	0.60	0.92	0.91	0.97

6.1.1 Performance in the two dimensional case

The main advantage of the test of symmetry by random projections is its easy implementation in infinite dimensions. Still, it is important to note its good performance in low dimensions. For instance when considering bivariate alternatives $X = (X_1, X_2)$, as in Einmahl and Gan (2016), under the following alternative hypotheses

- H_1) (A): $X_1,\ X_2 \sim exp(1) - 1$; X_1,X_2 independent.
- H_1) (B): $X_1,\ X_2 \sim pareto(1) - 1$; X_1,X_2 independent.
- H_1) (C): $\tilde{\Theta} \sim U(0, 2\pi)$, $R|\tilde{\Theta} \sim exp(1)$ if $\tilde{\Theta} \leq 5\pi/4$, $R|\tilde{\Theta} \sim exp(10)$ if $\tilde{\Theta} > 5\pi/4$, $(X_1, X_2) = (R\cos(\tilde{\Theta}), R\sin(\tilde{\Theta}))$.

Table 4 Proportion of rejections under different alternatives

Distribution	$n = 100$			$n = 200$		
H_1	T_n	Ley	RPK_{10}	T_n	Ley	RPK_{10}
(A)	0.99	0.97	0.90	1.00	0.99	0.99
(B)	0.85	1.00	0.87	1.00	1.00	0.95
(C)	0.81	0.77	0.95	1.00	0.97	1.00
(D)	1.00	0.99	0.99	1.00	1.00	1.00
(E)	0.86	0.31	0.77	0.99	0.59	0.90

- H_1) (D): $X_1,\ X_2 \sim \chi_1^2 - 1$; X_1,X_2 independent.
- H_1) (E): $X_1 \sim N(0,1)$, $X_2 \sim exp(1) - 1$; X_1, X_2 independent.

These alternative hypotheses are used to analyze the performance of the test in the continuous bivariate case. The test implemented in Einmahl and Gan (2016) rejects the null hypothesis for large values of the following statistic denoted by T_n. Let $(\tilde{\Theta}, R)$ be the polar coordinates of the vector (X_1, X_2), define

$$\Theta = \begin{cases} \tilde{\Theta} & \text{if } \tilde{\Theta} \in [0, \pi) \\ \tilde{\Theta} - \pi & \text{if } \tilde{\Theta} \in [\pi, 2\pi), \end{cases}$$

$$\delta = \begin{cases} 1 & \text{if } \tilde{\Theta} \in [0, \pi) \\ -1 & \text{if } \tilde{\Theta} \in [\pi, 2\pi), \end{cases}$$

$$U_i = F_{R|\Theta}(R_i|\Theta_i), \qquad \hat{U}_i = \hat{F}_{R|\Theta}(R_i|\Theta_i),$$

where $F_{R|\Theta}$ stands for the conditional distribution of $R|\Theta$ and $\hat{F}_{R|\Theta}$ is an estimator of $F_{R|\Theta}$. Then T_n is given by

$$T_n = \int_0^1 \int_0^\pi \int_{\theta_1}^\pi \left(\hat{W}_n(\theta_2, u) - \hat{W}_n(\theta_1, u)\right)^2 d\hat{G}(\theta_2) d\hat{G}(\theta_1) du$$
$$+ \int_0^1 \int_0^\pi \int_0^{\theta_1} \left(\hat{W}_n(\pi, u) - \hat{W}_n(\theta_1, u) - \hat{W}_n(\theta_2, u)\right)^2 d\hat{G}(\theta_2) d\hat{G}(\theta_1) du,$$

where $\hat{W}_n(\theta, u) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \delta_i \mathbb{I}_{[0,\theta]\times[0,u]}\left(\Theta_i, \hat{U}_i\right)$, and $\hat{G}$ stands for the empirical distribution of $\Theta_1, \ldots, \Theta_n$. The significance level is $\alpha = 0.05$ and 2000 replications have been performed. The sample sizes are $n = 100$ and $n = 200$. The results can be seen in Table 4.

6.2 Simulations for the independence test

The efficiency of the proposed test is measured through simulations. The competitor test under consideration, introduced by Székely et al. (2007), is relevant in the study

of the independence of random vectors. This test was adjusted by Székely and Rizzo (2013) to perform better in high dimensions. It is called the Covariance Distance test.

For complex functions γ on $\mathbb{R}^p \times \mathbb{R}^q$, the norm $\|\cdot\|_w$ is defined by

$$\|\gamma(s,t)\|_w^2 = \int_{\mathbb{R}^{p+q}} |\gamma(s,t)|^2 w(t,s)dtds,$$

where $w(t,s)$ is an arbitrary positive weight function. Given a pair of random vectors $(X,Y) \in \mathbb{R}^p \times \mathbb{R}^q$, Székely et al. (2007) define a measurement of dependence between random vectors called the *covariance distance*, given by

$$\begin{aligned} &\mathcal{V}(X,Y;w) = \|f_{X,Y}(t,s) - f_X(t)f_Y(s)\|_w^2, \\ &\mathcal{V}(X;w) = \mathcal{V}(X,X;w), \end{aligned} \tag{20}$$

where $f_X(t)$, $f_Y(t)$ stand for the corresponding characteristic functions of the random vectors X and Y and $f_{X,Y}$ stands for the joint characteristic function. In addition, they introduced a standardized version that is called *correlation distance*, given by

$$\mathcal{R}^2(X,Y) = \frac{\mathcal{V}(X,Y;w)}{\sqrt{\mathcal{V}(X;w)\mathcal{V}(Y;w)}}, \tag{21}$$

where

$$\|\cdot\|_w^2 = \int_{\mathbb{R}^{p+q}} |\cdot|^2 w(t,s)dtds.$$

The function $w(t,s)$ is a weight function, appropriately chosen

$$w(t,s) = (c_p c_q |t|_p^{1+p} |s|_q^{1+q})^{-1}, \quad \text{with} c_d = \frac{\pi^{(1+d)/2}}{\Gamma\left((1+d)/2\right)}. \tag{22}$$

Székely and Rizzo (2013) modify the statistic and determine their limit distribution under the independence of X and Y. This test is implemented for the *R* language in the *energy* package.

Three scenarios are used:

– (Scenario 1.) A sample $(X^{(k)}, Y^{(k)})$ of size 20 of random vectors X and Y is simulated with independent distributions, each one with marginal uniform distribution and Gumbel's copula with $\delta = 2$. The bivariate Gumbel copula is given by

$$C(x,y;\delta) = \exp\left(-\left[(-\log x)^\delta + (-\log y)^\delta\right]^{1/\delta}\right) \quad \delta \geq 1,$$

see Fig. 3.

There is a search for dependent alternatives replacing the last j coordinates of the vector Y with the first j coordinates of the vector X for different values of k, $k \in \{0, 5, 9, 13, 17, 21, 25, 29\}$.

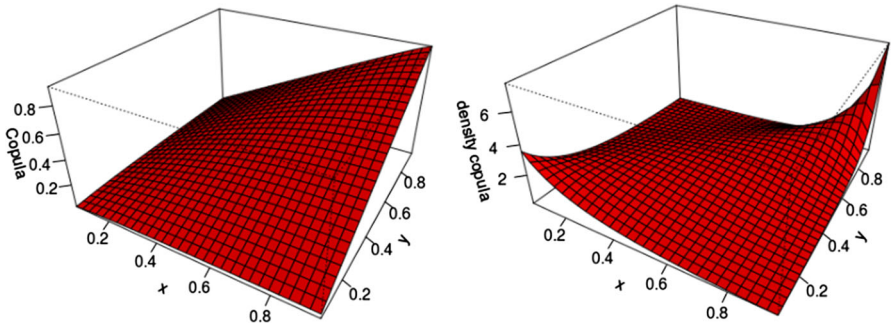

Fig. 3 For $\delta = 2$. *Left panel* Gumbel copula. *Right panel* the corresponding density function

- (Scenario 2.) A sample $(X^{(k)}, Y^{(k)})$ of size 20, of random vectors X and Y of dimension 50, each vector with independent distribution, each one with uniform marginals and Gumbel's copula with $\delta = 2$, is simulated. There is a search for dependent alternatives replacing the coordinates of the vector Y with the coordinates of the vector X for the coordinates of Y that exceed a certain threshold, denoted by u, which varies between 0 and 1.
- (Scenario 3.) A sample of size 20, of random vectors U, V and Z of dimension 50 is simulated. Each vector has independent distribution, each one with uniform marginal distribution and Gumbel's copula with $\delta = 2$. Let $X = (1 - \epsilon)U + \epsilon Z$ and $Y = (1 - \epsilon)V + \epsilon Z$ with $\epsilon \in [0, 1]$. $\epsilon = 0$ corresponds to the independent case.

The performance of the two competitors for the three scenarios will now be compared.

- (RPK.INDEP$_{50}$ Test.) For each data sample, 50 independent random directions h and 50 independent random directions f that are uniformly distributed on the unit sphere were generated. The samples of $\mathbf{X}$ and $\mathbf{Y}$ were projected onto each random direction h_j and f_j, $j = 1, \ldots, 50$ respectively. The p-value of the test given in Genest et al. (2007) was computed for each pair of projected samples onto the 50 random pairs of directions in $\mathbb{R}^2$. If at least in one of these directions H_0 was rejected, the null assumption was rejected. These procedures were performed $10,000$ times, and the proportion of p values smaller than $\alpha = 0.001$ for each univariate test was computed using Bonferroni correction.
- (DIST.COV. Test) The test is developed in Székely and Rizzo (2013), ($\alpha = 0.05$).

As shown in Table 5, in Scenarios 2 and 3 the random projections test using the RPK.INDEP$_{50}$ test in $\mathbb{R}^2$ behaves similarly to DIST.COV. test. In Scenario 1 our test (RPK.INDEP$_{50}$) has a better performance.

7 Real data: neuronal activity in alcoholics

The database consists of the EEG waves registered from 88 patients. This is an example of functional data. There were two groups of subjects: alcoholic and control, 44

Table 5 Power of the independence tests in Scenarios 1, 2 and 3

Scenario 1			Scenario 2			Scenario 3		
k	RPK.INDEP$_{50}$	DIST.COV	u	RPK.INDEP$_{50}$	DIST.COV	ϵ	RPK.INDEP$_{50}$	DIST.COV
0	0.01	0.05	1	0.04	0.04	0.00	0.04	0.08
5	0.10	0.12	0.86	0.19	0.21	0.14	0.05	0.11
9	0.55	0.13	0.71	0.60	0.51	0.29	0.10	0.12
13	0.82	0.24	0.57	0.85	0.87	0.43	0.21	0.31
17	0.98	0.26	0.43	0.96	0.97	0.57	0.72	0.92
21	1.00	0.63	0.26	0.98	0.99	0.71	1.00	1.00
25	1.00	0.78	0.14	1.00	1.00	0.86	1.00	1.00
29	1.00	0.93	0	1.00	1.00	1.00	1.00	1.00

subjects at each group. The objective is to determine the loss of association or synchronization between the two hemispheres of the brain due to the consumption of alcohol, and the areas of the brain in which the loss was greater. The data set is published in the UCI data repository.[1]

First, we fixed a matching triple of electrodes, and then performed the following steps.

Selecting the data:

- Select the samples for one subject for the three electrodes of the triple.
- Group these samples into three groups, one for each electrode. There are 256 measures for the same second for each electrode.
- For the three groups, there have been measured 60 different seconds.
- Thus, there are 60 vectors of 256 measures for each electrode.

Projecting the triple:

- For each node, we assume that the 60 trajectories in the sample are observed on 256 measures per second. This allows us, with regard to a random choice of $h \in E^*$, to generate the standard Brownian motion from a discrete version which exploits the independent-increments property, i.e. we start at 0 at time zero, and define iteratively the value at the next time step by adding an independent $N(0, 1/256)$ variable.
- Thus, we obtain 60 real values for each node. So we have a sample of size 60 in $\mathbb{R}^3$, see Fig. 5, where $\mathbf{X}_i = (X_{1,i}, X_{2,i}, X_{3,i})$ with $i = 1, 2, \ldots, 60$.

[1] https://archive.ics.uci.edu/ml/datasets/EEG+Database. A detailed description of the data set is available in Zhang et al. (1995). The database contains measurements from 64 electrodes that were placed on the subjects' scalps and sampled at 256 Hz (3.9-msec epoch) for 1 sec. In this study, 9 triples of nodes were used; 27 of the 64 electrodes are shown in Fig. 4. Each subject was exposed to either a single stimulus (S1) or two stimuli (S1 and S2). The stimuli were pictures of objects chosen from a picture set. Each observation has 256 measures of one second, and the id of the subject, the group and the sample are labeled. The EEG emission values are expressed in micro volts. By removing noise spectral decomposition using the fast Fourier transform of the signals was performed. The association between the different triples is studied as shown in Fig. 4. To simplify the procedure, the remaining signals were not processed.

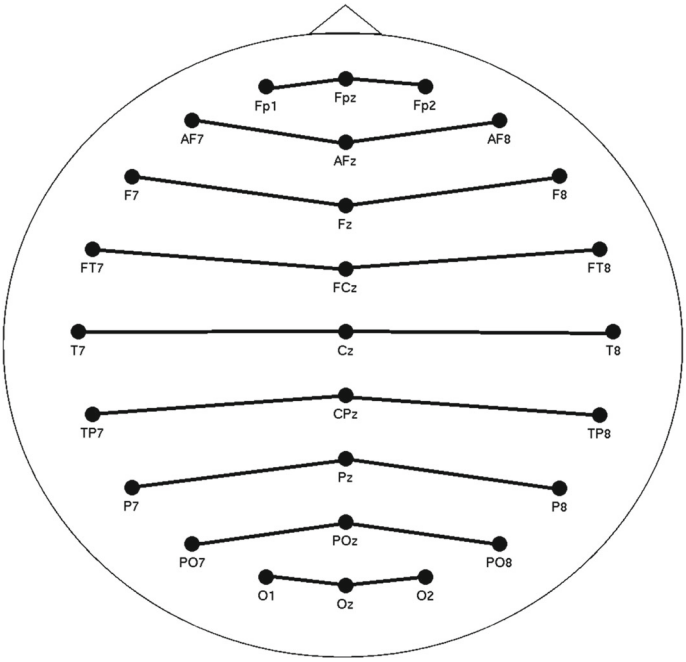

Fig. 4 Associated electrodes in the independence tests

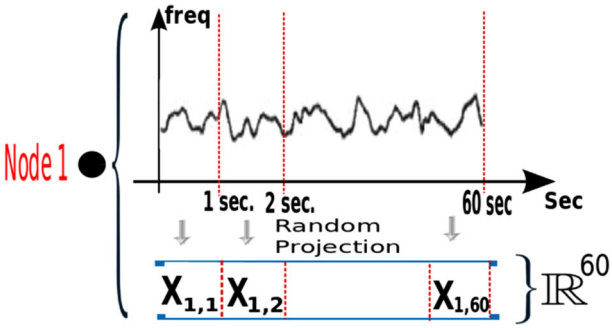

Fig. 5 Scheme at a node of the sample construction by random projections

Computing the p values for the triple:

– Using the new test, one obtains four p values: one for each pair and one for the triple.

This process is repeated for this triple for the 44 alcoholic subjects and the 44 control subjects. Thus, one obtains a set of 44×4 p values for the alcoholic subjects and the same for the control subjects.

Comparing the results:

– Make four pairs of boxplots.
– Each pair has a white plot for the alcoholics and a grey plot for the control subjects.

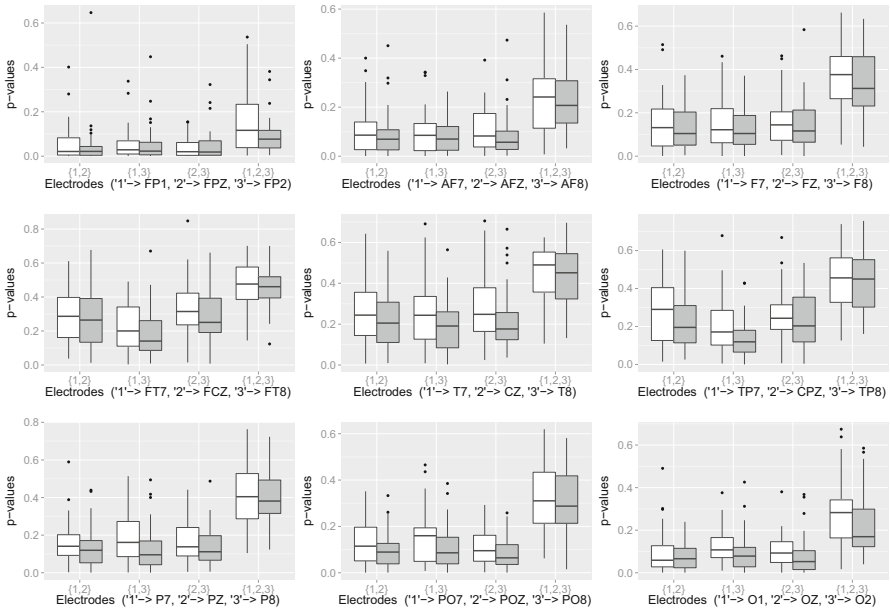

Fig. 6 Box diagrams of the means of the p values of the 50 independence tests of the signals as shown in Fig. 4, performed for 44 alcoholic patients (*white boxes*) and 44 control patients (*grey boxes*)

– Each box plot is generated from the 44 p values for each subset of electrodes of the triple (three pairs and one triple).

Then, the same process is performed for the other triples.

The results are shown in Fig. 6. There are two boxplots for each of the four sets of p values. The white box is for the alcoholics and the grey box is for the control subjects.

In neurodegenerative diseases such as Alzheimers disease (AD), three major effects on the EEG have been observed: slowing of the EEG, reduced complexity of the EEG signals, and perturbations in the synchrony of the EEG, Dauwels et al. (2010). Dawels reviews various measures of EEG synchrony considered in the context of AD diagnosis. The p value of the independence random projection-based test can be used as an association or synchrony measure. A greater p value indicates a greater loss of association.

As shown in Fig. 6, for most of the boxplots, the white ones (corresponding to the alcoholics sample) have a larger median and larger dispersion, which may indicate that the non–alcoholic sample has a better association. Also, for both populations the frontal area of the brain exhibits less loss of association than the rest.

8 Conclusions

– Two tests were developed based on random projections: a central symmetry test and an independence test. They are valid for finite dimensional data and functional

data taking values in a Banach space. They demonstrated an improved efficiency compared to other competitors in a small simulation study.
- Both tests are simply implemented, with rapid computational performance.
- In the case of the central symmetry test a proposal has been developed that is more general than those for sphericity and ellipticity of the data.
- Regarding the independence test it is possible to measure the association of several random elements, as shown in the example of EEG signals. The loss of synchronicity between the signals of opposite hemispheres of the brain is indicated in the case of alcoholic patients, and these symptoms are more visible under more complex stimuli.

Acknowledgements The authors gratefully acknowledge the constructive comments of the referees which help to improve the quality of the paper significantly. This work was partially supported by grant ID2014-48, CSIC, Udelar.

Appendix.

Proof of Theorem 3. Let M and Q be the probability measures induced by the random elements $\mathbf{X}$ and $-\mathbf{X}$. Then

$$\begin{aligned}\mathcal{E}(M, Q) &= \{h \in E^*/Q_h = M_h\}\\ &= \{h \in E^*/f(\mathbf{X})\text{and} - f(\mathbf{X})\text{have the same distribution}\}\\ &= \mathcal{E}(\mathbf{X}).\end{aligned}$$

From Theorem 2, because $\mathcal{E}(M, Q)$ has a positive μ-measure, it can be concluded that $Q = M$, namely, $\mathbf{X}$ and $-\mathbf{X}$ have the same distribution, so $\mathbf{X}$ is a centrally symmetric random element.

Proof of Theorem 4. Given two probability measures M and Q on $E \times E$ over the determinant class of the half-spaces $A = \{(x, y) \in E \times E / h(x, y) \leq t, h \in (E \times E)^*\}$,

$$\begin{aligned}M(A) &= \int \mathbf{1}_{\{h(x,y)\leq t\}} dP_{\mathbf{X}}(x) dP_{\mathbf{Y}}(y), \quad \forall h \in (E \times E)^*, \quad \forall t \in \mathbb{R},\\ Q(A) &= \int \mathbf{1}_{\{h(x,y)\leq t\}} dP_{(\mathbf{X},\mathbf{Y})}(x, y), \quad \forall h \in (E \times E)^*, \quad \forall t \in \mathbb{R}.\end{aligned} \tag{23}$$

The maximum is used as the norm on $E \times E$, namely

$$\|(x, y)\| = \max\{\|x\|_E, \|y\|_E\}, \tag{24}$$

therefore,

$$\begin{aligned} m_M(n) &= \int \|(x,y)\|^n dP_{\mathbf{X}}(x) dP_{\mathbf{Y}}(y) \\ &\leq \left[\underbrace{\int \|x\|^n dP_{\mathbf{X}}(x)}_{m_{\mathbf{X}}(n)} + \underbrace{\int \|y\|^n dP_{\mathbf{Y}}(y)}_{m_{\mathbf{Y}}(n)} \right] \\ &\leq 2\max\{m_{\mathbf{X}}(n), m_{\mathbf{Y}}(n)\}. \end{aligned}$$

So,

$$\begin{aligned} m_M^{-1/n}(n) &\geq 2^{-1/n} \left[\max\{m_{\mathbf{X}}(n), m_{\mathbf{Y}}(n)\}\right]^{-1/n} \\ &\geq 2^{-1/n} \left[\min\{m_{\mathbf{X}}^{-1/n}(n), m_{\mathbf{Y}}^{-1/n}(n)\}\right]. \end{aligned}$$

Thus, the moments are finite, and Carleman's condition for the measure M holds.

Let the set $\mathcal{E}(M, Q) = \{h \in (E \times E)^* / Q_h = M_h\}$. Let $h \in \mathcal{E}(\mathbf{X}, \mathbf{Y})$. Now, setting $f(\mathbf{X}) = h(\mathbf{X}, 0)$ and $g(\mathbf{Y}) = h(0, \mathbf{Y})$, one has that $h(\mathbf{X}, \mathbf{Y}) = f(\mathbf{X}) + g(\mathbf{Y})$. Then,

$$\begin{aligned} Q_h\left((-\infty, t]\right) &= P\left(f(\mathbf{X}) + g(\mathbf{Y}) \leq t\right) \\ &= \int_{E\times E} \mathbf{1}_{\{f(x)+g(y)\leq t\}} dP_{(\mathbf{X},\mathbf{Y})}(x,y) \\ &= \int_{\mathbb{R}\times\mathbb{R}} \mathbf{1}_{\{u+v\leq t\}} dP_{(f(\mathbf{X}),g(\mathbf{Y}))}(u,v) \\ &= \int_{\mathbb{R}\times\mathbb{R}} \mathbf{1}_{\{u+v\leq t\}} dP_{f(\mathbf{X})}(u) dP_{g(\mathbf{Y})}(v) \\ &= M_h\left((-\infty, t]\right). \end{aligned}$$

Then, $h \in \mathcal{E}(M, Q)$, implying that $\mathcal{E}(M, Q)$ has a positive μ-measure. By Theorem 2, it follows that $M = Q$, and thus $\mathbf{X}$ and $\mathbf{Y}$ are independent. □

Proof of Theorem 5. Under H_0, since $\mathbf{X}_1$ is symmetric for any $h \in E^*$, Theorem 1 implies that $h(\mathbf{X}_1) \in \mathbb{R}$ is also symmetric and therefore it fulfills the condition given in (11) for the null assumption in the one dimensional case. Let $h(\mathbf{Y}_1) \geq h(\mathbf{Y}_2) \geq \ldots \geq h(\mathbf{Y}_n)$ be the order statistics (sorted from the largest to the smallest) of the absolute values $|h(\mathbf{X}_1|, |h(\mathbf{X}_2|, \ldots, |h(\mathbf{X}_n|$. Let $t_{n,i}^h = F_n(-h(\mathbf{Y}_i))$. Then, $0 \leq t_{n,1}^h \leq t_{n,2}^h \leq \ldots \leq t_{n,n}^h \leq F(0) = 1/2$. Because F is continuous, ties not occur a.s. and therefore,

$$0 < t_{n,1}^h < t_{n,2}^h < \ldots < t_{n,n}^h < F(0) < 1/2 \quad \text{a.s.}$$

Via the canonical transformation, one may define $V_n^h(t) = n^{1/2}[G_n^h(t) - t]$ with $0 < t < 1$, where $G_n^h(t) = \frac{1}{n}\sum_{i=1}^n \mathbf{1}_{[-\infty,t)}\left(F^h(X_i)\right)$, and define

$$\tilde{V}_n^h(t) = V_n^h(t^-) + V_n^h(1-t), \quad 0 \le t \le 1/2. \tag{25}$$

Then, $\tilde{V}_n^h(t)$ is a stochastic process defined on $(0, 1/2)$ having n jumps of 1 or -1 at the points $t_{n,1}^h, t_{n,2}^h, \ldots, t_{n,n}^h$. Let $p_{i,j} = P\left(h(\mathbf{Y}_{n-i+1}) = |h(\mathbf{X}_j|\right)$. Then

$$\begin{aligned} &P\left(h(\mathbf{Y}_{n-i+1}) \quad \text{matches with a positive value of} \quad h(\mathbf{X}_j)\right) \\ &= \sum_{j=1}^{n} p_{ij} P\left(h(\mathbf{X}_j) > 0 \Big/ Y_{n-i+1}^h = |h(\mathbf{X}_j)|\right) = 1/2 \sum_{j=1}^{n} p_{i,j} = 1/2. \end{aligned} \tag{26}$$

Let $sg(.)$ stand for the sign function, and $|\cdot|$ for the absolute value function. It is well known that the vectors

$$(sg\,(h(\mathbf{X}_1))\,, \ldots, sg\,(h(\mathbf{X}_n))) \text{ and } (|h(\mathbf{X}_1)|\,, \ldots, |h(\mathbf{X}_n)|)$$

are independent under the null assumption.

Then, the jumps of $n^{1/2}\tilde{V}_n^h(t)$ at $t_{n,1}^h, t_{n,2}^h, \ldots, t_{n,n}^h$ are independent. Therefore, under H_0, the distributions of the statistics $D_-^h(n)$ and $D_+^h(n)$ follow the distribution of the maximum of a symmetric random walk of n steps from the origin, and $nD^h(n)$ follows the distribution of the maximum of the absolute values of the random walk as can be seen in Takács (1967). □

Proof of Theorem 6. Write P for the distribution of X and Q for the distribution of $-X$. The set $\mathcal{E}(P, Q)$ has H-measure zero in $\mathbb{R}^d$, since if the H-measure were positive, then by Theorem 3 X and $-X$ would have the same distribution, which contradicts the hypothesis.

For each $h \in \mathcal{E}^c(P, Q)$, by Theorem 1, $h(\mathbf{X})$ is not symmetric, so if we define

$$\delta(F^h) = \sup_{x \ge 0} |F^h(x) + F^h(-x) - 1|,$$

it holds that

$$\delta(F^h) = \begin{cases} 0 & \text{if } F^h \in \mathcal{F}_0^h \\ > 0 & \text{if } F^h \in \mathcal{F}_1^h. \end{cases} \tag{27}$$

There exists, under H_1, $t_h \in \mathbb{R}$ such that

$$P\left(x \in \mathbb{R}^d / \langle x, h\rangle \le t_h\right) \ne Q\left(x \in \mathbb{R}^d / \langle x, h\rangle \le t_h\right), \tag{28}$$

namely,

$$F^h(t_h) + F^h(-t_h) - 1 \ne 0. \tag{29}$$

By the Glivenko–Cantelli theorem, $\sup_x |F_n^h(x) - F^h(x)| \xrightarrow{a.s} 0$, which entails

$$\begin{aligned} D_n^h &\geq |F_n^h(t_h) + F_n^h(-t_h) - 1| \\ &\geq |F^h(t_h) + F^h(-t_h) - 1| - |F_n^h(t_h) - F^h(t_h)| - |F_n^h(-t_h) - F^h(-t_h)| \\ &\geq \frac{\delta(F^h)}{2}, \end{aligned}$$

almost surely when $n \to +\infty$.

References

Aki S (1993) On nonparametric tests for symmetry in R^m. Ann Inst Stat Math 45:787–800

Albert P, Ratnasinghe D, Tangrea J, Wacholder S (2001) Limitations of the case-only design for identifying gene-environment interactions. Am J Epidemiol 154:687–693

Azzalini A, Valle AD (1996) The multivariate skew-normal distribution. Biometrika 83:715–726

Blough DK (1989) Multivariate symmetry via projection pursuit. Ann Inst of Stat Math 41:461–475

Brandwein A, Strawderman W (1991) Generalizations of James–Stein estimators under spherical symmetry. Ann Stat 19:1639–1650

Cramér H, Wold H (1936) Some theorems on distribution functions. J Lond Math Soc 11:290–294

Cuesta-Albertos JA, Fraiman R, Ransford T (2006) Random projections and goodness of fit tests in infinite-dimensional spaces. Bull Braz Math Soc 37:477–501

Cuesta-Albertos JA, Fraiman R, Ransford T (2007) A sharp form of the Cramer–Wold theorem. J Theor Prob 20:201–209

Cuevas A, Fraiman R (2009) On depth measures and dual statistics. A methodology for dealing with general data. J Multivariate Anal 100:753–766

Dauwels J, Vialatte F, Cichocki A (2010) Diagnosis of alzheimers disease from EEG signals: where are we standing. Curr Alzheimer Res 7:487–505

Dyckerhoff R, Ley C, Paindaveine D (2015) Depth-based runs test for bivariate central symmetry. Ann Inst Stat Math 67:917–941

Einmahl J, Gan Z (2016) Testing for central symmetry. J Stat Plann Inference 169:27–33

Fermaninan JD (2005) Goodness-of-fit tests for copulas. J Multivariate Anal 95:119–152

Fermaninan JD, Radulovic D, Wegkamp M (2004) Weak convergence of empirical copula processes. Bernoulli 10:847–860

Friedman JH, Stuetzle W (1981) Projection pursuit regression. J Am Stat Assoc 76:817–823

Friedman JH, Tukey JW (1974) A projection pursuit algorithm for exploratory data analysis. IEEE Trans Comput 23:881–890

Genest C, Quessy JF, Rémillard B (2007) Asymptotic local efficiency of Cramér-Von Mises tests for multivariate independence. Ann Stat 35:166–191

Genest C, Rémillard B (2008) Validity of the parametric bootstrap for goodness-of-fit testing in semiparametric models. Ann l'ins Henri Poincaré (B) Probab Stat 44:1096–1127

Hallin M, Paindaveine D (2002) Optimal tests for multivariate location based on interdirections and pseudo-Mahalanobis ranks. Ann Stat 30:1103–1133

Heathcote C, Rachev S, Cheng B (1995) Testing multivariate symmetry. J Multivariate Anal 54(1):91–112

Jones M, Sibson R (1987) What is projection pursuit? J R Stat Soc Ser A 150:1–36

Ley, C.: Univariate and multivariate symmetry: statistical inference and distributional aspects. Ph.D. thesis, Université libre de Bruxelles (2010)

Marden, J.: Multivariate analysis, design of experiments, and survey sampling, chap. 14. Multivariate rank test, pp. 401–432. CRC Press (1999)

Mason DM, Schuenemeyer JH (1983) A modified Kolmogorov–Smirnov test sensitive to tail alternatives. Ann Stat 11:933–946

Neuhaus G, Zhu L (1998) Permutation tests for reflected symmetry. J Multivariate Anal 67(2):129–153

Padgett WJ, Taylor RL (1973) Laws of large number for normed linear spaces and Certain Fréchet spaces. Springer, Berlin

Puri ML, Sen PK (1971) Nonparametric methods in multivariate analysis. Wiley, New York, p 440
Sen PK, Chatterjee SK (1973) On Kolmogorov-Smirnov type test for symmetry. Ann Inst Stat Math 25:288–300
Sen PK, Puri ML (1967) On the theory of rank order tests for location in the multivariate one sample problem. Ann Math Stat 38:1216–1228
Serfling, R.: Multivariate symmetry and asymmetry. In: Encyclopedia of Statistical Sciences, Second Edition, vol. 8, pp. 5338–5345. J. Wiley & Sons (2006)
Shohat JA, Tamarkin JD (1943) The problem of moments. Mathematical Surveys and Monographs
Székely GJ, Rizzo ML (2013) The distance correlation t-test of independence in high dimension. J Multivariate Anal 117:193–213
Székely GJ, Rizzo ML, Bakirov N (2007) Measuring and testing dependence by correlation of distances. Ann Stat 35:2769–2794
Takács L (1967) Combinatorial methods in the theory of stochastic processes. Wiley, Hoboken
Wilks SS (1935) On the independence of k sets of normally distributed statistical variables. Econometrica 3:309–326
Zhang X, Begleiter H, Porjesz B, Wang W, Litke A (1995) Event related potentials during object recognition tasks. Brain Res Bull 38(6):531–538